

List Coloring of Join of Paths

Hsu-Ling Liu and Justie Su-Tzu Juan*

Department of Computer Science and Information Engineering
National Chi Nan University, Puli, Nantou 54561, Taiwan
{s96321510, jsjuan}@ncnu.edu.tw

Abstract

A graph G is called L -list colorable if there is a coloring of vertices that the color assigned to a vertex v is from a list $L(v)$ and any two adjacent vertices have different colors. The smallest k such that G is L -list colorable for every assignment L satisfying $|L(v)| = k$ for all vertex v is called the list chromatic number of G and denoted by $\chi_l(G)$. We study the list chromatic number of the join of two graphs. The equation: $\chi_l(G_1 \vee G_2) = \chi_l(G_1) + \chi_l(G_2)$ for vertex-disjoint graphs G_1 and G_2 , is considered that may be true. However, the fact does not behave so simply. In this paper, we prove that $\chi_l(P_3 \vee P_n) = \chi_l(P_3) + \chi_l(P_n)$ if $1 \leq n \leq 15$. On the other hand, we construct an example of $P_3 \vee P_{24}$ to say the list chromatic number of $P_3 \vee P_{24}$ is ≥ 5 . And we prove $\chi_l(P_3 \vee P_n) = \chi_l(P_3) + \chi_l(P_n) + 1$ if $n \geq 24$. That implies $\chi_l(P_m \vee P_n) \neq \chi_l(P_m) + \chi_l(P_n)$ for all $m \geq 3$ and $n \geq 24$.

Keywords: Join of two graphs, List chromatic number, List coloring.

1 Introduction

List coloring was introduced independently by Vizing [8] in 1976 and by Erdős, Rubin and Taylor [3] in 1979. Each vertex v of a graph G is assigned a color from a list $L(v)$ of allowed colors. G is called L -list colorable if there is a coloring c of vertices with:

- $c(u) \neq c(v)$ for $uv \in E$,
- $c(v) \in L(v)$ for all $v \in V$.

We say that G is k -choosable if it is L -list colorable for every assignment L satisfying $|L(v)| = k$

(also be denoted by k -list) for all vertex v . The list chromatic number of G (also called choice number, $ch(G)$), denoted by $\chi_l(G)$, is the smallest k such that G is k -choosable.

For any two graphs G_1, G_2 , the join of G_1 and G_2 , denoted by $G_1 \vee G_2$, is obtained by the union of G_1 and G_2 adding the edge uv with $u \in V_1$ and $v \in V_2$ for $G_i = (V_i, E_i)$, $i = 1, 2$. Figure 1 is an example to present the join operation.

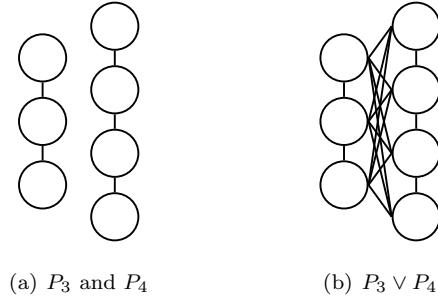


Figure 1: The join of P_3 and P_4 .

In [2, 4, 9], the equation: $\chi_l(G_1 \vee G_2) = \chi_l(G_1) + \chi_l(G_2)$, was mentioned that does not behave so simply. There exists an example in [5] to show that $\chi_l(G_1 \vee G_2)$ are not always equal to $\chi_l(G_1) + \chi_l(G_2)$ for vertex-disjoint graphs G_1, G_2 . They proved that $\chi_l(K_{m,n}) = m + 1$ if $n \geq m^m$. However, $\chi_l(K_m^c) + \chi_l(K_n^c) = 2$ for K_i^c is an empty graph with i vertices. And, it is easy to see that $m + 1 \neq 2$ for $m \neq 1$. Therefore, in this paper, we will discuss the list chromatic number restricted to $P_m \vee P_n$ if m is specific vertex number.

Figure 2 is an example for list coloring of $P_3 \vee P_4$. In Figure 2(a), every vertex v has a 4-list $L(v)$. And, there exists a list coloring presented at Figure 2(b) such that the graph is colored properly.

The following are some notations that will be used in this paper. The parameters $\Delta(G)$ and $\delta(G)$ are the maximum and the minimum degree

*Corresponding author.

of G , respectively. The smallest number of colors needed to assign every vertex of G such that no two adjacent vertices share the same color is called *chromatic number*, $\chi(G)$. The *coloring number* $\text{col}(G)$ of G is the smallest integer d for which there exists an ordering $v_1, v_2, \dots, v_n$ of the vertices of G such that each vertex v_i has at most $d-1$ neighbors among the vertices $v_1, \dots, v_{i-1}$ (see [1]). In [7], Tuza gave the following inequalities for every graph G

$$\chi(G) \leq \chi_l(G) \leq \text{col}(G) \leq \Delta(G) + 1. \quad (1)$$

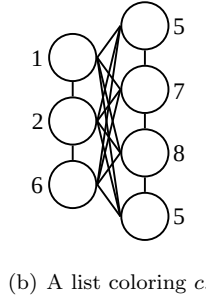
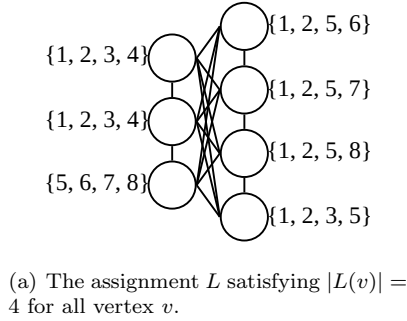


Figure 2: An example of a list coloring c in $P_3 \vee P_4$.

2 Main Results

A simple lower bound of the list chromatic number restricted to the join of two paths will show in Subsection 2.1. Subsection 2.2 will give the list chromatic numbers of $P_1 \vee P_n$ and $P_2 \vee P_n$ for any n . Meanwhile, this subsection will also give an upper bound for the list chromatic number of the join of P_3 and P_n for any n . Subsection 2.3 will show $P_3 \vee P_{24}$ is not 4-choosable and conclude by giving the exact value of $\chi_l(P_3 \vee P_n)$ for $n < 16$ or $n \geq 24$.

2.1 Lower Bound

The following theorem is a simple lower bound for the list chromatic number of the join of paths.

Theorem 1. *Let P_m and P_n be two paths with m and n vertices respectively. Then, the list chromatic number of the join of P_m and P_n ,*

$$\chi_l(P_m \vee P_n) \geq \begin{cases} 2, & \text{if } m = n = 1; \\ 3, & \text{if } m = 1 \text{ exclusive-or } n = 1; \\ 4, & \text{otherwise.} \end{cases}$$

Proof. In [6], Stiebitz and Škrekovski showed that $\chi(G_1 \vee G_2) = \chi(G_1) + \chi(G_2)$ for any graphs G_1 and G_2 . Hence, by inequality (1), we have

$$\begin{aligned} \chi_l(P_m \vee P_n) &\geq \chi(P_m \vee P_n) \\ &= \chi(P_m) + \chi(P_n). \end{aligned}$$

If $m = n = 1$, $\chi_l(P_m \vee P_n) \geq 1 + 1 = 2$. If only one of m and n equals 1, say $m = 1$ and $n \geq 2$, without loss of generality. Then, $\chi_l(P_m \vee P_n) \geq 1 + 2 = 3$. Otherwise, $m \geq 2$ and $n \geq 2$, hence $\chi_l(P_m \vee P_n) \geq 2 + 2 = 4$. $\square$

2.2 Upper Bound

First, we will present the list chromatic number of path, $P_1 \vee P_n$ and $P_2 \vee P_n$ respectively. Then, we will give an upper bound for the list coloring restricted to $P_3 \vee P_n$.

Lemma 2. *For $n \geq 2$, the list chromatic number of the path with n vertices is 2.*

Proof. For $n \geq 2$, it is easy to see $\chi_l(P_n) \geq \chi(P_n) = 2$. Let $P_n = (v_1, v_2, \dots, v_n)$ be a path which has n vertices, and give any 2-list for every vertex. Choose any color in $L(v_1)$ to be $c(v_1)$, then for $i = 2, 3, \dots, n$, color v_i with $c(v_i) \in L(v_i)$ and $c(v_i) \neq c(v_{i-1})$. Since $|L(v_i) - c(v_{i-1})| \geq 1$, it must exist such $c(v_i)$. In other words, every vertex has at most one colored neighbor, and then it can be assigned one color from its list. Therefore, $\chi_l(P_n) = 2$ if $n \geq 2$. $\square$

Theorem 3. *Let P_n be a path with n vertices. Then,*

$$\chi_l(P_1 \vee P_n) = \begin{cases} 2, & \text{if } n = 1; \\ 3, & \text{if } n \neq 1. \end{cases}$$

Proof. According to the vertex number of P_n , we have the following two cases.

- $n = 1$.

Because $P_1 \vee P_1 \cong P_2$, by Lemma 2, $\chi_l(P_1 \vee P_1) = 2$.

- $n \neq 1$.

Let $V(P_1) = \{u\}$, $V(P_n) = \{v_1, v_2, \dots, v_n\}$ and $E(P_n) = \{v_i v_{i+1} | 1 \leq i \leq n-1\}$ with $n \geq 2$. Given any 3-list to every vertex, and then assign a color $c(u) \in L(u)$ to u . Therefore, the available colors of $L(v_i) - c(u)$ are at least two. In other words, the vertex v_i are colored properly from the list $L(v_i) - c(u)$ with $i = 1, 2, \dots, n$ by Lemma 2. For $n \neq 1$, $P_1 \vee P_n$ is 3-choosable. Hence, $\chi_l(P_1 \vee P_n) = 3$ by Theorem 1. $\square$

Theorem 4. *Let P_n be a path with n vertices. The list chromatic number of $P_2 \vee P_n$ is 4 if $n \geq 2$.*

Proof. Let $V(P_2) = \{u_1, u_2\}$ and $V(P_n) = \{v_1, v_2, \dots, v_n\}$ with $n \geq 2$; $E(P_1) = \{u_1 u_2\}$ and $E(P_n) = \{v_i v_{i+1} | 1 \leq i \leq n-1\}$. Given any 4-list, and then color the vertices u_1, u_2, v_1, v_2 from their lists respectively. Because $|L(v_3) - \{c(u_1), c(u_2), c(v_2)\}| \geq 1$, the vertex v_3 can be colored. In the same way, the vertex v_i can be colored because $|L(v_i) - \{c(u_1), c(u_2), c(v_{i-1})\}| \geq 1$ for $4 \leq i \leq n$. That is, along this order $u_1, u_2, v_1, v_2, \dots, v_n$ to color, $P_2 \vee P_n$ can be colored properly. For the above reasons and Theorem 1, $\chi_l(P_2 \vee P_n) = 4$ if $n \geq 2$. $\square$

Theorem 6 mainly studies the list chromatic number of the graph $G = P_3 \vee P_n$ for $n \geq 3$. It contains two upper bounds of $\chi_l(G)$ and obtains the exact value of list chromatic number for $n < 16$. In the proof of this theorem, we use Lemma 5 frequently. So, we have to see the lemma first.

Lemma 5. *Let P_n be a path with n vertices. If at most one vertex in P_n has 1-list and others have 2-list, then P_n is L -list colorable.*

Proof. Let $P_n = (v_1, v_2, \dots, v_n)$ and denoted the vertex with 1-list by $v_{i'}$ for $i' \in \{1, 2, \dots, n\}$. If $i' = 1$ (or $i' = n$), i.e. $v_{i'}$ is an endpoint of P_n , we can assign a color to each vertex in this order: $v_1, v_2, \dots, v_n$ (or $v_n, v_{n-1}, \dots, v_1$, respectively). Because $|L(v_p) - c(v_{p-1})| \geq 1$ (or $|L(v_{p-1}) - c(v_p)|$, respectively) for $p = 2, 3, \dots, n$, P_n is colored properly. If i' is not an endpoint of P_n , choose $c(v_{i'}) \in L(v_{i'})$. Next, color the other vertices in order $v_{i'+1}, v_{i'+2}, \dots, v_n, v_{i'-1}, v_{i'-2}, \dots, v_1$. Since $|L(v_p) - c(v_{p-1})| \geq 1$ for $p = i' + 1, i' + 2, \dots, n$ and $|L(v_p) - c(v_{p+1})| \geq 1$ for $p = i' - 1, i' - 2, \dots, 1$, we can assign a proper color $c(v_p) \in L(v_p)$ to the

vertex v_p for each $1 \leq p \leq n$. Therefore, P_n is L -list colorable $\square$

Theorem 6. *For $n \geq 3$, the join of P_3 and P_n is 5-choosable. Moreover, the list chromatic number of $P_3 \vee P_n$ is 4 if $n < 16$.*

Proof. Let $V(P_3) = \{v_1, v_2, v_3\}$, $E(P_3) = \{v_1 v_2, v_2 v_3\}$, $V(P_n) = \{v_4, v_5, \dots, v_{n+3}\}$ and $E(P_n) = \{v_i v_{i+1} | 4 \leq i \leq n+2\}$. Because $\delta(G) = \min\{3, n\} + 1 = 4$ and there exists an order of the vertices of G , $(v_1, v_2, \dots, v_{n+3})$, such that each vertex v_i has at most $\delta(G)$ neighbors among the vertices $v_1, \dots, v_{i-1}$, for $i = 2, 3, \dots, n+3$. So, $\chi_l(G) \leq \text{col}(G) = \delta(G) + 1 = 5$ by (1).

For the vertex number n of P_n that does not exceed 16, we have another upper bound 4 by the following cases, that equals the lower bound in Theorem 1. So, $\chi_l(P_3 \vee P_n) = 4$ if $n < 16$. Now, given any 4-list $L(v)$ to set every vertex v of G .

- Case 1. $L(v_1) \cap L(v_3) \neq \emptyset$.

In this case, we will consider the number of the available elements in the list of each vertex that belongs to P_n if the vertices of P_3 are colored. Choose color $c(v_1) = c(v_3)$ from $L(v_1) \cap L(v_3)$ to vertices v_1 and v_3 . Then, there exists a color $c(v_2)$ that set to v_2 because $|L(v_2) - \{c(v_1), c(v_3)\}| \geq 3$. That is, $|L(v_j) - \{c(v_1), c(v_2), c(v_3)\}| \geq 2$ for $j = 4, 5, \dots, n+3$. Therefore, the vertices of G can be colored properly by Lemma 2.

- Case 2. $L(v_1) \cap L(v_3) = \emptyset$.

We claim that when $n = 15$, there exists one way to choose $c(v_i) \in L(v_i)$ for $1 \leq i \leq n+3 = 18$ such that c is a proper coloring.

Let $|L(v_1) \cap L(v_2)| = k_{12}$ and $|L(v_2) \cap L(v_3)| = k_{23}$, then $0 \leq k_{12}, k_{23} \leq 4$ and $0 \leq k_{12} + k_{23} \leq 4$ because $L(v_1) \cap L(v_3) = \emptyset$. For $i = 1, 2, 3$, there are $4^3 - 4 \times (k_{12} + k_{23})$ ways to choose $c(v_i)$. Hence, there exist at least $64 - 4 \times 4 = 48$ ways to choose $c(v_i)$. However, when we consider $\{c(v_1), c(v_2), c(v_3)\}$, some ways are regarded as the same in these 48 ways. For example, one way includes $c(v_1) = a, c(v_2) = b, c(v_3) = c$, and another way includes $c(v_1) = b, c(v_2) = a, c(v_3) = c$. The two ways are the same because $\{a, b, c\} = \{b, a, c\}$. Therefore, the minimum number of different $\{c(v_1), c(v_2), c(v_3)\}$ in these 48 ways is 24 when $k_{12} = 4, k_{23} = 0$ (or $k_{12} = 0, k_{23} = 4$). Except for 24, the

number of other ways must be ≥ 36 to determine $\{c(v_1), c(v_2), c(v_3)\}$. According to the number of different $\{c(v_1), c(v_2), c(v_3)\}$, there are two sub-cases in the following.

Case 2.1. The number of different $\{c(v_1), c(v_2), c(v_3)\}$ is ≥ 36 .

Each $L(v_j)$ contains $\binom{4}{3} = 4$ subsets for $j = 4, 5, \dots, n+3$ such that $L(v_j)$ contains at most $15 \times 4 = 60$ subsets which elements are 3 for all $4 \leq j \leq n+3$. Since $36 \times 2 = 72 > 60$, there must exist one way to choose $c(v_i) \in L(v_i)$ for $i = 1, 2, 3$ such that $\{c(v_1), c(v_2), c(v_3)\}$ is contained by at most one $L(v_{j'})$ for $j' \in \{4, 5, \dots, n+3\}$. Because $|L(v_{j'}) - \{c(v_1), c(v_2), c(v_3)\}| \geq 1$ and $|L(v_p) - \{c(v_1), c(v_2), c(v_3)\}| \geq 2$ for $j' \in \{4, 5, \dots, n+3\}, p = 4, 5, \dots, j'-1, j'+1, j'+2, \dots, n+3$, P_n is L -list colorable by Lemma 5, i.e. $P_3 \vee P_n$ is 4-choosable for $n < 16$.

Case 2.2. The number of different $\{c(v_1), c(v_2), c(v_3)\}$ is 24.

The case occurs at $k_{12} = 4, k_{23} = 0$ (or $k_{12} = 0, k_{23} = 4$). Without loss of generality, say $L(v_1) = L(v_2) = \{1, 2, 3, 4\}$, $L(v_3) = \{5, 6, 7, 8\}$. For any $j = 4, 5, \dots, n+3$, we only need to consider the cases that $L(v_j) \subset L(v_1) \cup L(v_2) \cup L(v_3)$, $L(v_j) \neq L(v_1)$ ($= L(v_2)$) and $L(v_j) \neq L(v_3)$. If not, $L(v_j)$ is always colored properly for $j = 1, 2, \dots, n+3$ by Lemma 5. The α - β distribution means that α elements of the 4-list are from $L(v_1) = L(v_2) = \{1, 2, 3, 4\}$ and other elements are from $L(v_3) = \{5, 6, 7, 8\}$ for $(\alpha, \beta) = (3, 1)$ or $(2, 2)$ or $(1, 3)$. So, the 4-list of each vertex v_j is one of 3-1, 2-2 or 1-3 distributions. For $4 \leq j \leq n+3$, let the number of vertex v_j with list which is 3-1, 2-2 or 1-3 distributions be x , y or z respectively. For $j = 4, 5, \dots, n+3$, every $L(v_j)$ which is α - β distribution contains $\binom{\alpha}{2} \binom{\beta}{1}$ 3-subsets, where 3-subset means a subset of $L(v_j)$ with 3 elements. It implies that $L(v_j)$ contains $\binom{3}{2} \binom{1}{1} x + \binom{2}{2} \binom{2}{1} y + \binom{1}{2} \binom{3}{1} z = 3x + 2y + 0z$ 3-subsets in all.

If each different $\{c(v_1), c(v_2), c(v_3)\}$ of these 24 ways is contained by exactly two lists of vertices in P_n . Then, we have

$$\begin{cases} x + y + z = 15 \\ 3x + 2y + 0z \geq 24 \times 2 = 48 \end{cases}$$

It implies that $x - 2z \geq 18$, i.e. $x \geq 18$ because $z \geq 0$. It contradict $0 \leq x \leq 15$. So, there must exist one way

to choose $c(v_1), c(v_2), c(v_3)$ such that there is at most one vertex $v_{j'}$ in P_n with $|L(v_{j'}) - \{c(v_1), c(v_2), c(v_3)\}| \geq 1$. Since other vertex $v_p \in V(P_n)$ with $|L(v_p) - \{c(v_1), c(v_2), c(v_3)\}| \geq 2$ for $p = 4, 5, \dots, j'-1, j'+1, j'+2, \dots, n+3$, P_n is L -list colorable by Lemma 5, i.e. $P_3 \vee P_n$ is 4-choosable for $n < 16$. $\square$

2.3 An Example

In this subsection, we give an example as Figure 3, $P_3 \vee P_{24}$, and show it is not 4-choosable. If it is 4-choosable, there are 6 ways to color u_1 and u_2 as follows.

- Case 1. $c(u_1) = 1$ and $c(u_2) = 2$.

If $c(u_3) = 5$, the vertex v_1 must be assigned color 3 and $L(v_2) - \{c(u_1), c(u_2), c(u_3), c(v_1)\} = \emptyset$, i.e. v_2 cannot choose available color from its list. No matter which color u_3 is assigned, there exist two vertices v_i and v_{i+1} for $i = 1, 3, 5, 7$ such that these two vertices cannot be colored properly.

- Case 2. $c(u_1) = 1$ and $c(u_2) = 3$.

Similarly to Case 1. When u_3 is assigned a color from $L(u_3)$, there exist one vertex of v_i and v_{i+1} , for $i = 1, 3, 5, 7$, that can not be colored properly.

- Case 3. $c(u_1) = 1$ and $c(u_2) = 4$.

Assign u_3 the color 5 from $L(u_3)$. Then, $c(v_9)$ must be colored 3 and $L(v_{10}) - \{c(u_1), c(u_2), c(u_3), c(v_9)\} = \emptyset$. Hence, there exist one vertex v_i , for $i = 9, 10, \dots, 16$, that cannot be colored properly if u_3 is assigned any color from its list.

- Case 4. $c(u_1) = 2$ and $c(u_2) = 3$.

Let $c(u_3) = 5$, and then $c(v_{17}) = 4 \in L(v_{17}) - \{c(u_1), c(u_2), c(u_3)\}$. It implies that $L(v_{18}) - \{c(u_1), c(u_2), c(u_3), c(v_{17})\} = \emptyset$. Therefore, there is a vertex v_i for $i = 17, 18, \dots, 24$ such that no available color can be assigned if u_3 is set a color.

- Case 5. $c(u_1) = 2$ and $c(u_2) = 4$.

The reason is similar to Case 4. We can find a vertex v_i for $i = 17, 18, \dots, 24$ which will be colored improperly.

- Case 6. $c(u_1) = 3$ and $c(u_2) = 4$.

Assign any color from $L(u_3)$ to the vertex u_3 . Similarly to Case 3, there are two vertices v_i

and v_{i+1} for $i = 9, 11, 13, 15$ such that one is set the last available color and the other cannot be colored properly.

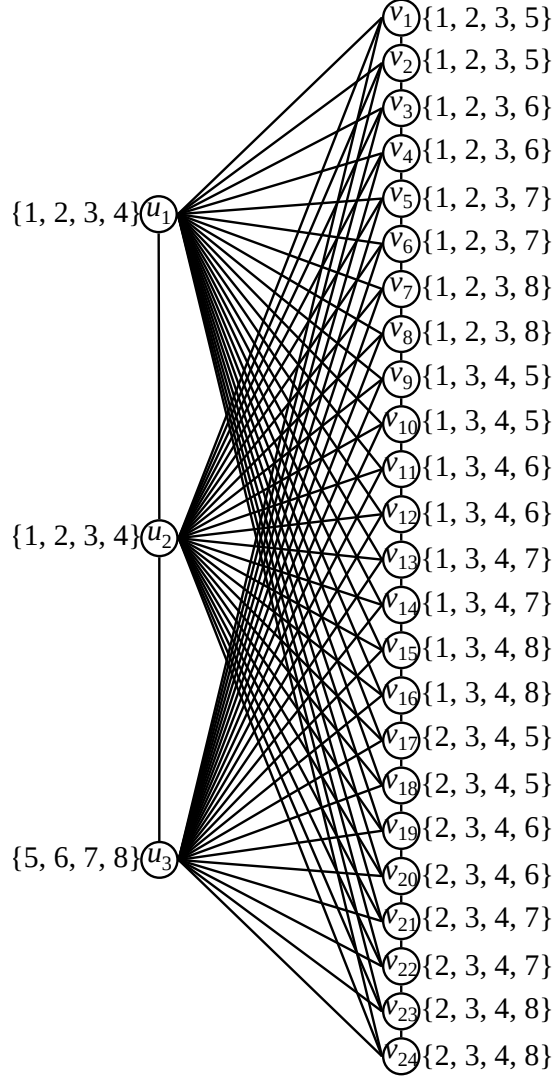


Figure 3: The graph $P_3 \vee P_{24}$ with 4-list of each vertex.

Property 7. Let H be a subgraph of graph G , then $\chi_l(H) \leq \chi_l(G)$.

According to the above example, we know that $P_3 \vee P_{24}$ is not 4-choosable, i.e. $\chi_l(P_3 \vee P_{24}) \geq 5$. Using Property 7, it implies that $\chi_l(P_3 \vee P_n) \geq 5$ since $P_3 \vee P_{24} \subseteq P_3 \vee P_n$ for all $n \geq 24$. And, by Theorem 3, 4 and 6, we have the following corollary.

Corollary 8. Let G be the join of P_3 and P_n , then

$$\chi_l(G) = \begin{cases} 3, & \text{if } n = 1; \\ 4, & \text{if } 2 \leq n < 16; \\ 5, & \text{if } n \geq 24. \end{cases}$$

3 Conclusion

In this paper, we present the list chromatic number of $P_m \vee P_n$ for any n and $m \leq 3$. If $2 \leq n < 16$, $P_3 \vee P_n$ is 4-choosable; if $n \geq 24$, it is 5-choosable. Hence, $\chi_l(P_3 \vee P_n)$ does not always equal $\chi_l(P_3) + \chi_l(P_n) = 4$. Moreover, using the result and Property 7, since $P_3 \vee P_n \subseteq P_m \vee P_n$ for $m \geq 3$, we have a better lower bound for the join of longer paths.

Corollary 9. Let P_m and P_n be two paths with m and n vertices respectively. For $m \geq 3$ and $n \geq 24$, the list chromatic number of $P_m \vee P_n$ is equal to or more than 5.

Also, if graph G is a *traceable* graph, i.e. G contains a Hamiltonian path, we have another related result.

Corollary 10. If G_1 and G_2 are traceable graphs with $|V(G_1)| \geq 3$, then

$$\chi_l(G_1 \vee G_2) \geq \begin{cases} 4, & \text{if } |V(G_2)| < 16; \\ 5, & \text{if } |V(G_2)| \geq 24. \end{cases}$$

Further research directions include that find the list chromatic number of $P_3 \vee P_n$ for $16 \leq n < 24$. Actually, we conjecture that $\chi_l(P_3 \vee P_n) = 4$ for $16 \leq n < 24$. And extend the number of vertices in P_3 to any number m and consider the exact value of $\chi_l(P_m \vee P_n)$ is another future work.

References

- [1] M. Borowiecki, S. Jendrol, D. Král, and J. Miškuf, “List coloring of Cartesian products of graphs,” *Discrete Mathematics*, Vol. 306, pp. 1955–1958, 2006.
- [2] S. Dantas, S. Gravier, and F. Maffray, “Extremal graphs for the list-coloring version of a theorem of Nordhaus and Gaddum,” *Discrete Applied Mathematics*, Vol. 141, pp. 93–101, 2004.
- [3] P. Erdős, A. L. Rubin, and H. Taylor, “Choosability in graphs,” *Congressus Numerantium*, Vol. 26, pp. 125–157, 1979.

- [4] S. Gravier, F. Maffray, and B. Mohar, “On a list-coloring problem,” *Discrete Mathematics*, Vol. 268, pp. 303–308, 2003.
- [5] D. G. Hoffman and P. D. J. Jr., “On the choice number of $K_{m,n}$,” *Congressus Numerantium*, Vol. 98, pp. 105–111, 1993.
- [6] M. Stiebitz and R. Škrekovski, “A map colour theorem for the union of graphs,” *Journal of Combinatorial Theory Ser. B*, Vol. 96, pp. 20–37, 2006.
- [7] Z. Tuza, “Graph colorings with local constraints—A survey,” *Math. Graph Theory*, Vol. 17, pp. 161–228, 1997.
- [8] V. G. Vizing, “Coloring the vertices of a graph in prescribed colors (in russian),” *Metody Diskret. Anal.*, Vol. 29, pp. 3–10, 1976.
- [9] B. Wu and L. Zhang, “Extension of a list coloring problem,” *Applied Mathematics Letters*, Vol. 19, pp. 135–139, 2006.